

THE POSTS AND TELECOMMUNICATIONS INSTITUTE OF TECHNOLOGY

DEPARTMENT OF INFORMATION TECHNOLOGY 1



PROJECT REPORT

MODULE 24

NAME: NGUYỄN TUẤN MINH

STUDENT ID: B22DCKH077

CLASS: E22HTTT

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I. Requirement

1. Glossary list

#	Name	Meaning
Related to human		
1	Staff	A person that works in the restaurant and have an account to use the system
2	Management Staff	A person with the highest authority including view statistics, manage dish information, make combo menus
3	Warehouse Staff	A person responsible for import ingredients from suppliers, manage supplier information
4	Sale Staff	A person who receives customers, takes orders, receives payments at the table, makes membership cards for customers, confirms table reservation information and order online
5	Customer	A person who benefits from the service, can search, book a table and order food online
6	Supplier	Partner providing ingredients
Related to activity		
7	View statistics	An action of manager staff that viewing the information of dish (name, description, price), ingredient (name, price, stock), customer (name, phone, email), supplier (name, phone, address)
8	Manage dish information	An action of manager staff that including adding new dish, changing or deleting existed dish information (name, description, price) in database
9	Make combo menu	An action of manager staff that combining two or more dish into a combo menu (name, description, list of dishes, price) and insert its information into database
10	Import ingredients from suppliers	An action of warehouse staff that confirming all the ingredients brought from suppliers and insert new ingredient or update stock of the existed ingredients in database
11	Manage supplier information	An action of warehouse staff that adding new supplier information (name, phone, email) and changing or deleting existed supplier information in database
12	Receive customer	An action of sale staff that meeting customer at reception hall, checking customer's information (name, phone, email) and guild them
13	Take orders	An action of sale staff that taking note of customer's orders (list of dishes, beverages) face-to-face and inform to chefs

14	Receive payment at the table	An action of sale staff that receiving customer's payment after they finish their meal and print bill (id, name, list of dishes and combos, price for each and total)
15	Make membership card for customer	An action of sale staff that creating and issue a card for customer to earn loyal points for their next reservation
16	Confirm table reservation information	An action of the sale staff that must be done when the customer comes to the restaurant. The customer will tell the sale staff about their name, phone number, the table they pre-booked and show the staff their mail that the restaurant sent to confirm their successful reservation. The sale staff will check their information and lead them to the table pre-booked.
17	Order online	An action of sale staff. The customer first calls the restaurant to book a table. The sale staff picks up and then checks the available table according to the schedule the customer gives them then informs the customer. If the customer agrees to the staff suggestion, the staff will ask the customer's information to make a table reservation and save it to the database. After that, the sale staff will send a mail to confirm their successful reservation
18	Add new dish	An action of management staff that is included in managing dish information action. The management staff will select adding a new dish option in the app interface then fill the dish information in (name, description, price). The dish information will be saved into database
19	Search table	An action of customer to search for available table in a specific time online
Related to object		
20	Dish	A specific type of food that has been prepared and is ready to be served or eaten. Its information will be saved into database (id, name, description, price)
21	Ingredient	A specific food item is used to make a dish. For example, in a salad, the ingredients are lettuce, tomatoes, cucumbers, and dressing. In a cake, the ingredients are flour, sugar, eggs, and milk. Its information will be saved into database (id, name, price)

22	Combo	Short for "combination," is a package deal that bundles several individual items together and sells them as a single offer, usually at a discounted price. In the context of a restaurant, a combo typically includes a main dish, a side dish, and a drink. Its information will be saved into database (id, name, list of dishes and beverages, price)
23	Menu	A list of the food and beverages that are available for purchase. The customers can see the menu online or menu book on the table. It contains name of the dish, price
24	Order	A request made by a customer to a restaurant for food and drinks. Customer can make order right at the table or via website (order online)
25	Payment	It is the process of a customer settling the cost of their order. The sale staff will receive money from customer and update payment information in database
26	Table	A physical location in the restaurant where customers are seated and is linked to a customer order
27	Membership card	A physical or digital card issued by the restaurant to its customers. It is linked to a customer's account in the system and grants them special privileges (such as discounts, loyalty points)

2. Project information

a. Object

A website system for restaurant management including viewing statistics (dishes, ingredients, customers, suppliers), managing dish information, making combo menus for manager; importing ingredients from suppliers, managing suppliers information for warehouse staff; receiving customers (taking orders, receiving payments at the table, make membership cards, confirm table reservation information) for sale staff. Booking table and order foods online for customer

b. Scope

- Staff:
 - + Login/ logout
- Manager:
 - + View statistics
 - + Manage dish information
 - + Making combo menus
- Warehouse staff:
 - + Import ingredients from suppliers
 - + Manage supplier's information
- Sale staff:
 - + Confirm table reservation
 - + Taking order
 - + Receiving payment

- + Make membership card for customer
- Customer:
 - + Booking table
 - + Order foods

c. How the module works

Management staff adds new dish: Staff logs into the system → selects the menu to manage dish information → selects the function of adding dish information → the system displays adding dish information interface with text fields (name, description, price) → enters dish information and clicks add → The system saves to the database.

Management staff views customer statistics: Selects the menu to view statistical reports → selects customer statistics by revenue → the system displays an interface with a combo box to select start and end date → selects the time to start and end statistics → view with a list of customers (name, phone number, revenue) → selects a customer to view expenses details at chosen start and end date → the system displays an interface with the order list that the customer has called (date, price) → selects view 1 order → views detail the selected order (list of dishes with number and price).

d. Information about objects

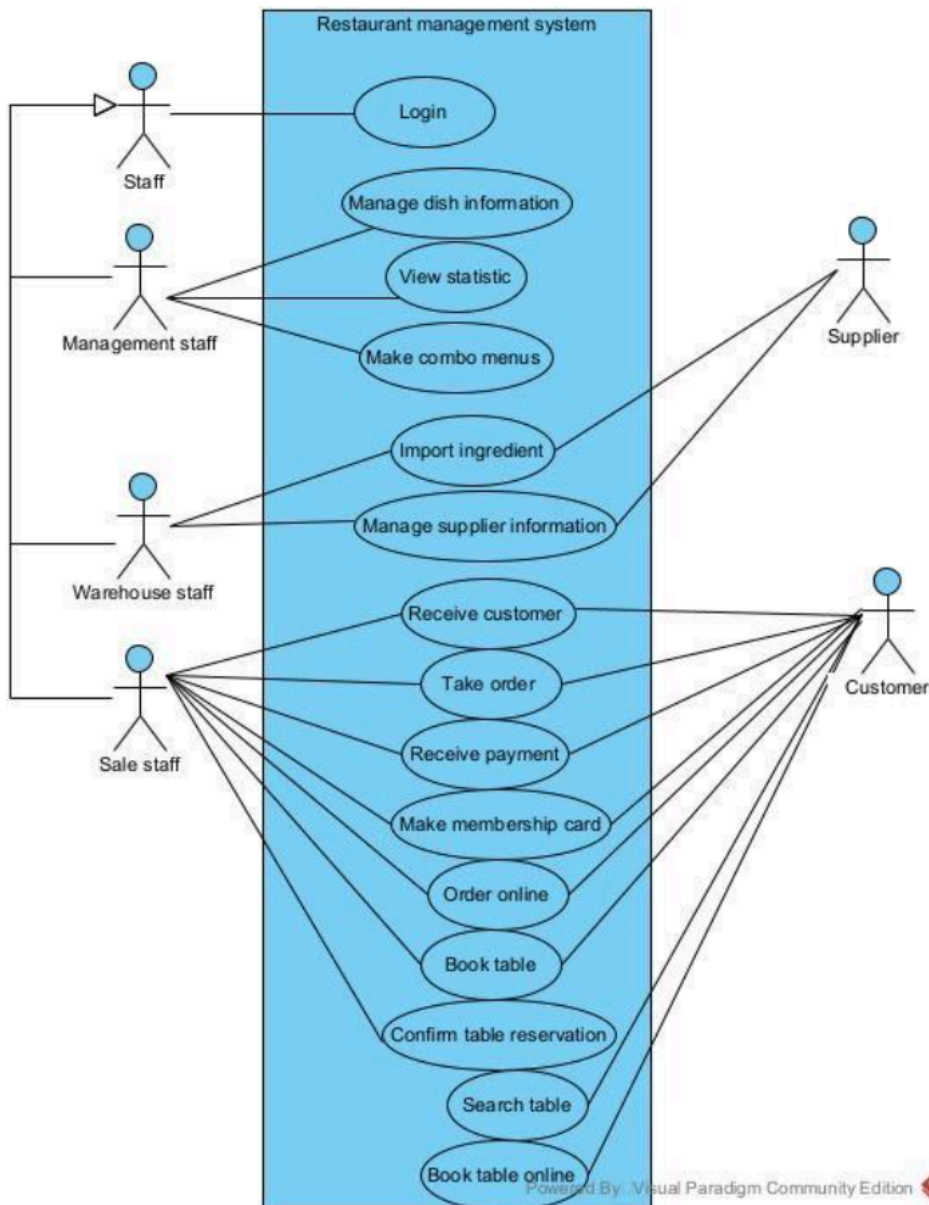
- Staff: username, password, name, phone number, email, role
- Customer: name, phone number, email
- Reservation: checkin, checkout, status
- Invoice: Paymentdate, status
- Dish: name, description, price
- Order: date, price
- Table: type, available
- Supplier: name, phone, email
- Ingredient: name, stock, price
- Combo: name, price
- Membership: type, point, start date, due date
- Import invoice: import date, price

e. Relationships among objects

- Customer – Membership: 1 – 1
- Customer – Reservation: 1 – n
- Reservation – Order: 1 – n
- Reservation – Table: n – 1
- Reservation – Invoice: 1 – n
- Order – Invoice: n – 1
- Order – Dish: n – n
- Order – Combo: n – n
- Dish – Combo: n – n
- Sale staff – Invoice: 1 – n
- Ingredient – ImportInvoice: n – n
- Supplier – Import invoice: 1 – n
- Warehouse staff – Import invoice: 1 – n

3. Use case diagram

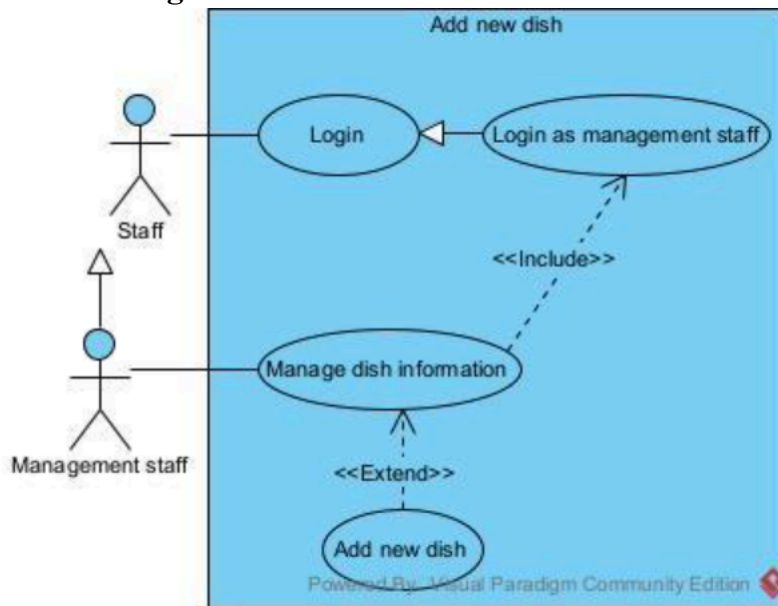
a. General use case



- Description:
 - + Login: This use case enables the staff to login into the restaurant system
 - + Manage dish information: This use case enables the management staff to adding new dish or change/delete exited dish
 - + View statistic: This use case enables the management staff to view the information of dish, ingredient, customer, supplier
 - + Make combo menus: This use case enables the management staff to make combo menus
 - + Import ingredient: This use case enables the warehouse staff to import ingredient
 - + Manage supplier information: This use case enables the warehouse staff to adding new supplier information or change/delete exited one
 - + Receive payment: This use case enables the sale staff to receiving customer's payment after they finish their meal and print bill
 - + Take order: This use case enables the sale staff to receive order offline

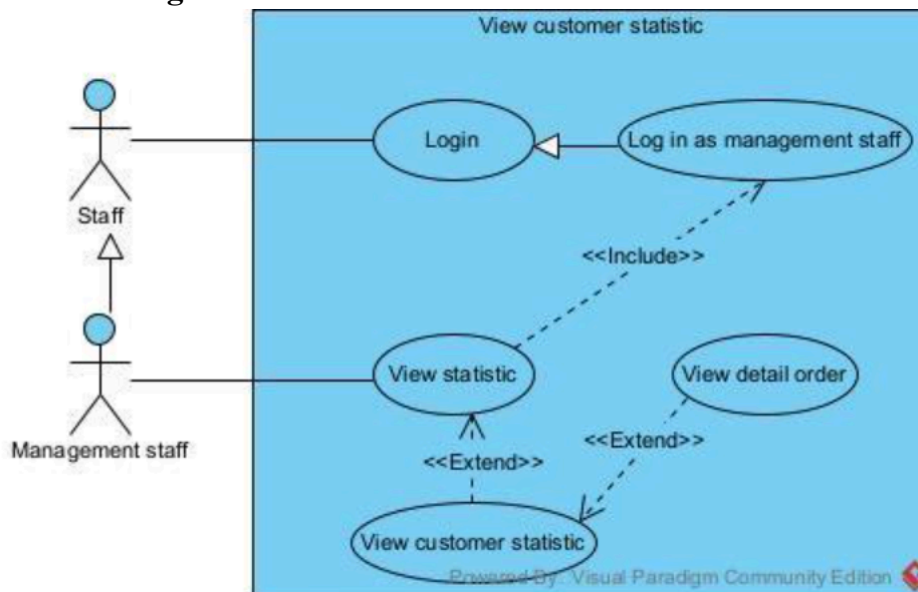
- + Confirm table reservation: This use case enables the sale staff to confirm the customer table reservation
- + Make membership card: This use case enables the sale staff to create and issue a card for customer to earn loyal points for their next reservation
- + Order online: This use case enables the customer to order online
- + Book table: This use case enables the sale staff to book table offline according to customer's request
- + Confirm reservation: This use case enables the sale staff confirm the reservation of the customer
- + Search table: This use case enables the customer to search for available table
- + Book table online: This use case enables the customer to book table online

b. Management staff adds new dish use case



- Login as management staff: This use case enables the management staff to login into their account
- Add new dish: This use case enables the management staff to add new dish into database

c. Management staff views customer statistic



- View customer statistic: This use case enables the management staff to view customer statistic
- View detail order: This use case enables the management staff to view a detail order in the list of orders of the selected customer

II. Analysis

1. Scenario

a. Module 1

Use case	Add new dish			
Actor	Management staff			
Pre-condition	The management staff has an account			
Post-condition	The management staff has added new dish into database			
Main events	1. The management staff logins with username = “a”, password = “12345” into the system to add new dish			
	2. The main interface which contains a Manage dish information button appears			
	3. The management staff selects the Manage dish information button			
	4. The system displays manage dish interface with an Add new dish button and a list of all dishes:			
	#	Name	Description	Price
	1	b	A delicious meal	100.000
	5. The management staff click the Add new dish button			
	6. The system shows the Add new dish interface with input of dish name, description, price and an Add button			
	7. The management staff enters dish name = “Noodle”, description = “a staple food made from unleavened dough that is rolled flat and cut, or extruded, into long strips or various other shapes”, price = “10.000” and clicks Add button			
	8. The system displays the success message box			
9. The staff clicks OK button				
10. The system returns to the manage dish interface				
#	Name	Description	Price	
1	b	A delicious meal	100.000	
2	Noodle	a staple food made from unleavened dough that is rolled flat and cut, or extruded, into long strips or various other shapes	10.000	
Exception	2. The system shows error message box			
	2.1. The management staff clicks OK button			
	2.2. The system returns to the login interface			

b. Module 2

Use case	View customer statistics
Actor	Management staff
Pre-condition	The management staff has logged in the account

Post-condition					
Main events	1. After login, the management staff selects View statistics button from the main interface				
	2. The system displays an interface that has 3 buttons which is View dish statistics, View customer statistics, View supplier statistic				
	3. The management clicks on View customer statistics				
	4. The system shows an interface with the input of start and end time of the statistics and a Search button				
	5. The management staff enters start time = “9/9/2025”, end time = “16/9/2025” and click Search				
	6. The system displays a list of customers				
	#	Name	Phone	Email	Revenue
	1	a	098720	A@gmail.com	1.000.000
	2	ab	213230	AB@gmail.com	500.000
	7. The management staff click on the first row of the list				
	8. The system displays the order list interface of the customer				
	#	Order id	Order date	Total amount	
	1	1	10/9/2025	600.000	
2	2	13/9/2025	400.000		
9. The management staff select the first row					
10. The system displays an interface contains order date = “10/9/2025”, Total amount= “600.000” and a list of dishes and combos that was ordered					
List of dishes					
#	Name	Price	Amount		
1	Noodle	10.000	10		
2	Korean dish	100.000	5		
List of combos					
#	Name	Price	Amount		
1	Family combo	50.000	2		
Exception	6. No customer displays on the list				

2. Entity class

a. Step 1:

The system supports the management of information about dishes, ingredients, customers, suppliers, orders, reservations, and combo menus in a restaurant. The system enables management staff to manage dish information, set up combo menus, manage suppliers, and view statistical reports such as dish statistics, customer statistics, and supplier statistics. The system enables warehouse staff to import ingredients from suppliers and manage supplier information. The system enables sales staff to receive customers, take orders, confirm reservations, process payments, and issue membership cards for customers. The system enables customers to search for dish information, book a table online, and order food online.

Once a payment or import is processed, the system generates invoices with details about the ordered dishes, imported ingredients, suppliers, and customers involved.

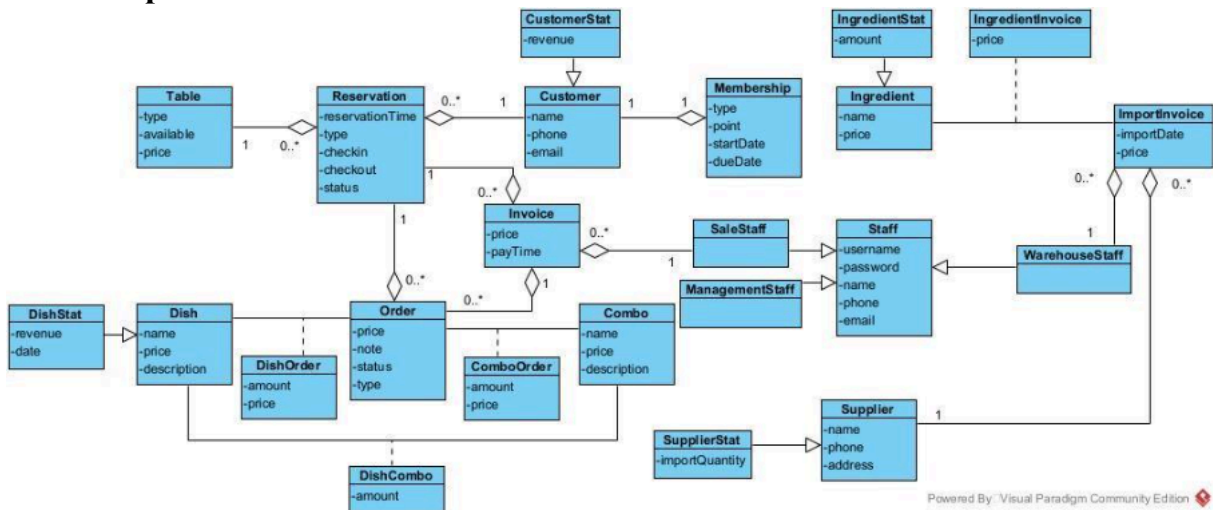
b. Step 2 + 3:

- Restaurant: Out of scope => eliminate
- System: Too abstract => eliminate
- Management staff, sale staff, warehouse staff => Class Staff: username, password, name, role
- Statistics: Too abstract => eliminate
- Dishes => Class Dish: name, price, description
- Ingredients => Class Ingredient: name, price, amount
- Customers => Class Customer: name, phone, email
- Suppliers => Class Supplier: name, phone, email
- Information: Too general
- Combo => Class Combo: name, price, description
- Menu: Too general
- Orders => Class Order: price, note, type, status
- Payment => Class Invoice: price, payTime
- Membership card => Class Membership: type, point, startDate, dueDate
- Table => Class Table: type, available
- Reservation => Class Reservation: reservationTime, type
- Food: Too general
- Invoice for imported ingredients => ImportInvoice: importDate, price

c. Step 4

- Customer – Membership: 1 – 1
- Customer – Reservation: 1 – n
- Reservation – Invoice: 1 – n
- Reservation – Order: 1 – n
- Reservation – Table: n – 1
- Order – Invoice: n – 1
- Order – Dish: n – n => create a new class DishOrder: amount, price
- Order – Combo: n – n => create a new class ComboOrder: amount, price
- Dish – Combo: n – n => create a new class DishCombo: amount
- Ingredient – ImportInvoice: n – n => create a new class IngredientInvoice
- Supplier – ImportInvoice: 1 – n
- WarehouseStaff – ImportInvoice: 1 – n
- SaleStaff – Invoice: 1 – n

d. Step 5

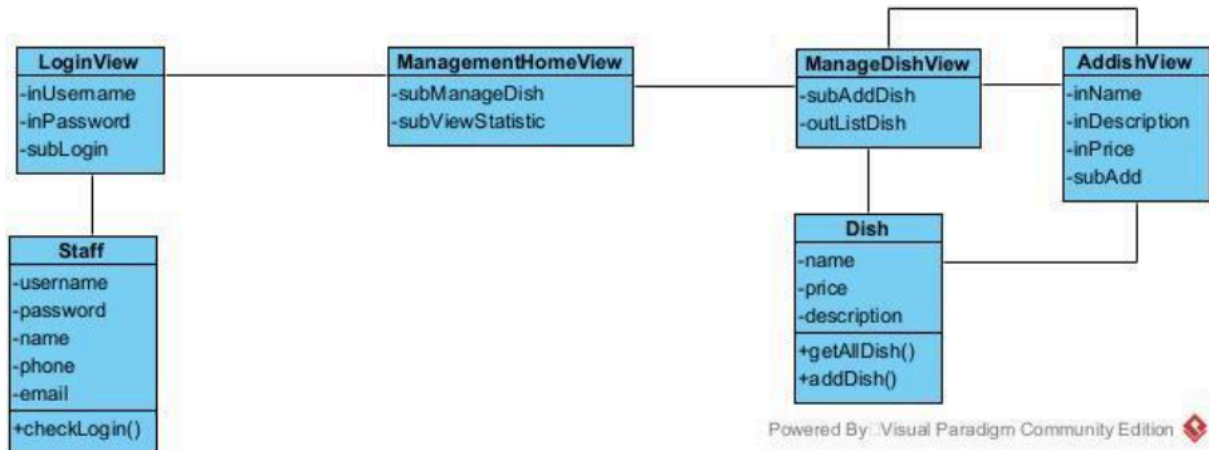


3. Module 1:

a. Class diagram

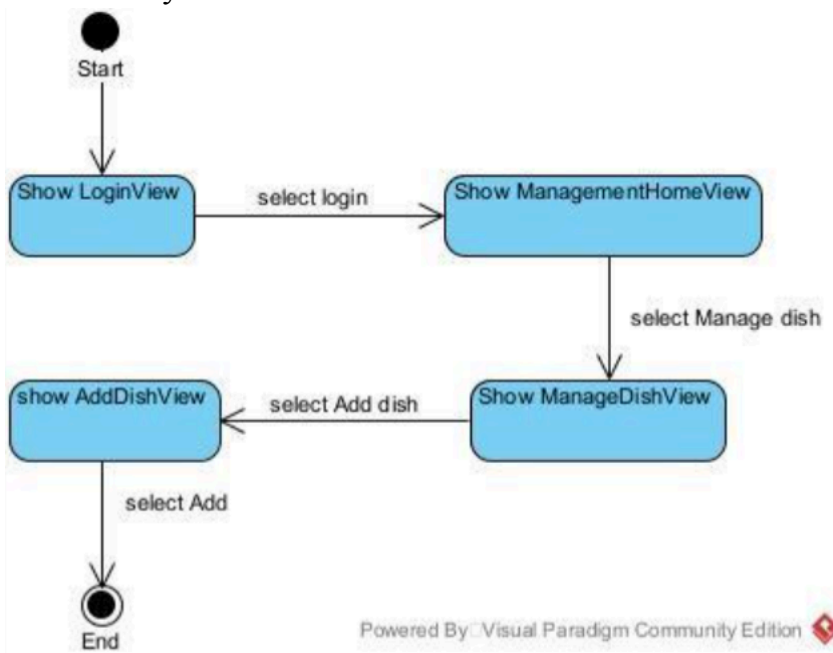
- Management staff logs into the system: need a class: LoginView:
 - + Input for username: inUsername
 - + Input for password: inPassword
 - + Login button: subLogin
- Enter the username/ password to login: need a method: checkLogin()
 - + Input: username, password (of the class Staff)
 - + Output: boolean
 - + Assign to class Staff
- Main interface contains Manage dish button appear => need a class: ManagementHomeView:
 - + Manage dish button => subManageDish
 - + View statistics => subViewStatistic
- System displays an interface with 2 buttons which is Edit dish and Add new dish => need a class: ManageDishView
 - + Add dish button => subAddDish
 - + A list of all dishes=> outListDish
- Display a list of all dishes => need a method: getAllDish():
 - + Input: None
 - + Output: List of Dish
 - + Assign to class Dish
- System shows the Add new dish interface => need a class: AddDishView
 - + Input for dish name: inName
 - + Input for dish description: inDescription
 - + Input for price description: inPrice

- + Add button: subAdd
- The dish is added into the database => need a method: addDish()
 - + Input: name, description, price (belongs to class Dish)
 - + Output: boolean
 - + Assign to class Dish



b. State diagram:

- From LoginView, after entering username/ password and select login, the system will move to ManagementHomeView
- From ManagementHomeView, select Manage Dish option will have the system move to ManageDishView
- From ManageDishView, when select option add dish, the system will show the AddDishView
- From AddDishView, after entering username/ description/ price and click Add, the system will save it into database and finish

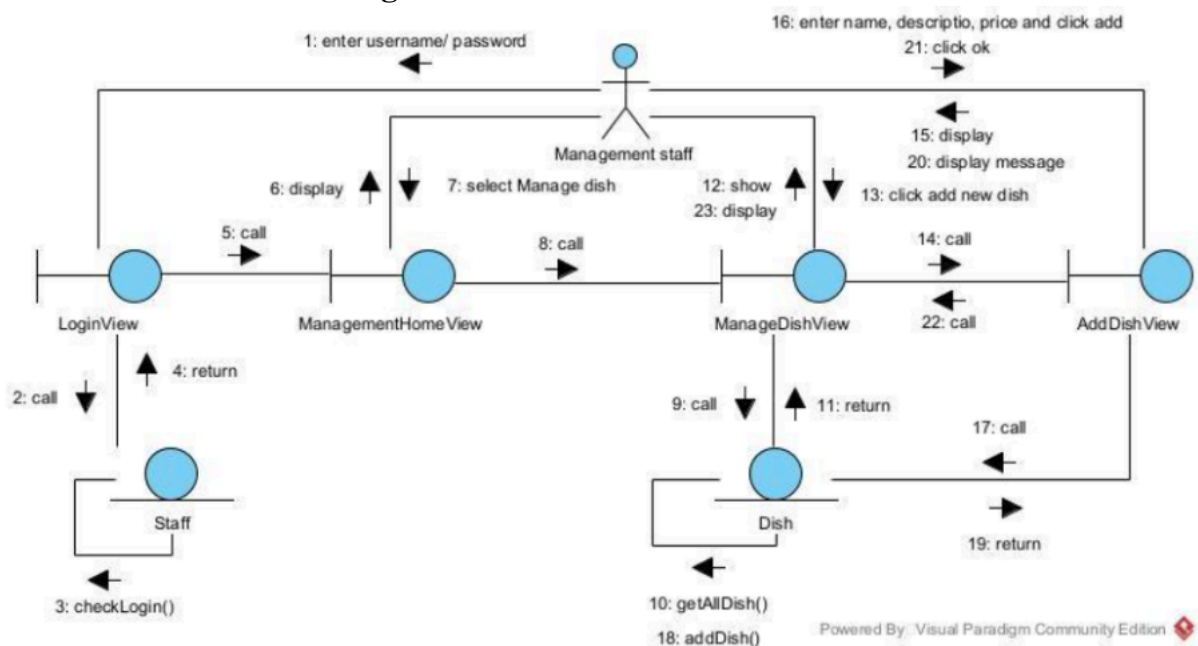


c. Scenario ver 2:

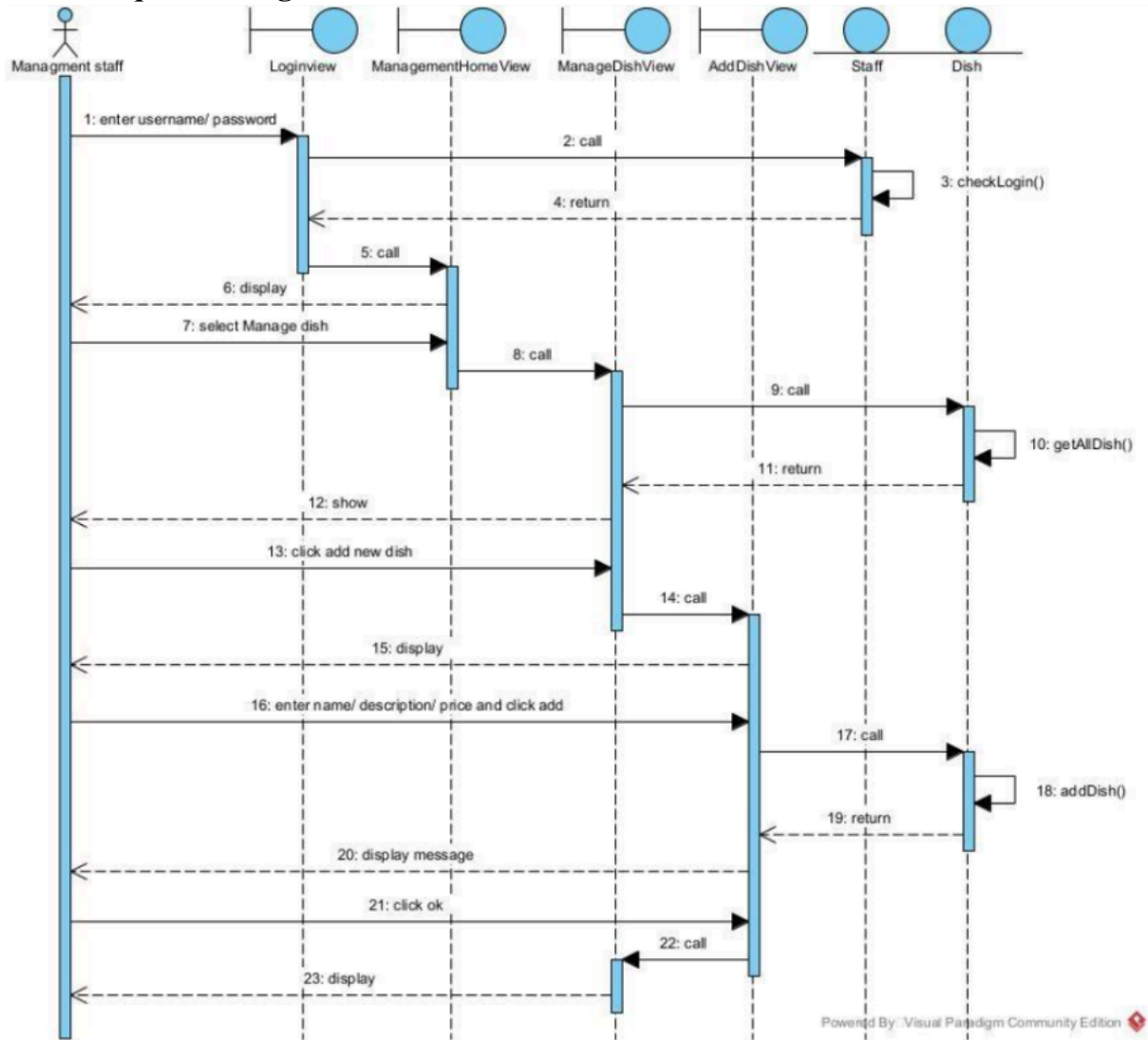
1. The management staff enters username/ password and clicks Login button
2. The class LoginView calls the class Staff to process

3. The class Staff calls the method checkLogin()
4. The class Staff returns the results to the class LoginView
5. The class LoginView calls the class ManagementHomeView
6. The class ManagementHomeView displays itself to the management staff
7. The management staff clicks on the Manage dish button
8. The class ManagementHomeView calls the class ManageDishView
9. The class ManageDishView calls the class Dish to process
10. The class Dish calls the method getAllDish()
11. The class Dish returns the result to the class ManageDishView
12. The class ManageDishView shows itself to the management staff
13. The management staff clicks on Add new dish button
14. The class ManageDishView calls the class AddDishView
15. The class AddDishView displays itself to the management staff
16. The management staff enters dish name, description, price and click Add button
17. The class AddDishView calls the class Dish to process
18. The class Dish calls the method addDish()
19. The class Dish returns to the class AddDishView
20. The class AddDishView displays a successful message to the management staff
21. The management staff clicks OK button of the message
22. The class AddDishView calls the class ManageDishView
23. The class ManageDishView displays itself to the management staff

d. Communication diagram



e. Sequence diagram:

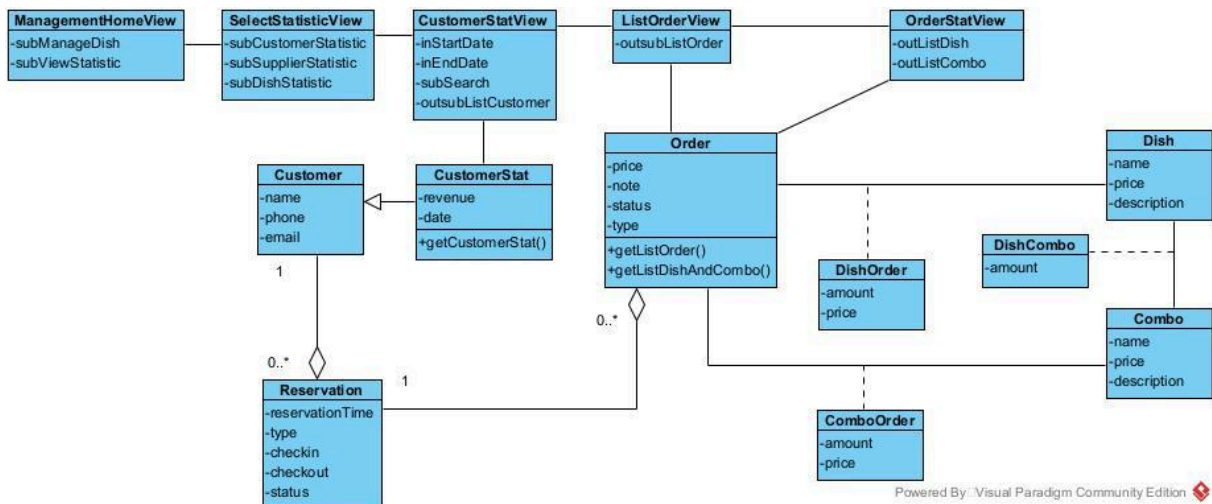


4. Module 2:

a. Class diagram

- Management staff selects View statistics from the main interface => need a class: ManagementHomeView:
 - + View statistic button: subViewStatistic
- System displays an interface that has 3 buttons which is View dish statistics, View customer statistics, View supplier statistic => need a class: SelectStatisticView:
 - + View customer statistics button: subCustomerStatistic
 - + View supplier statistics button: subSupplierStatistic
 - + View dish statistics button: subDishStatistic
- System shows an interface with the input of start and end time of the statistics and a Search button: need a class: CustomerStatView:
 - + input of start date: inStartDate
 - + input of end date: inEndDate
 - + Search button: subSearch

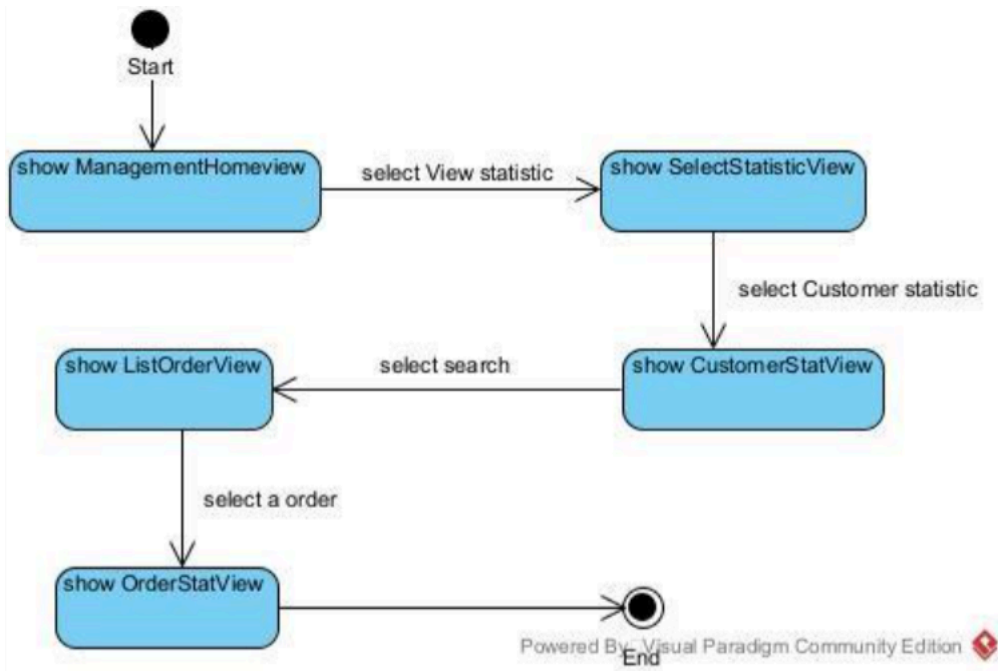
- + Display list of orders: outsubListCustomer
- Enter start and end time of the statistics and search => need a method: `getCustomerStat()`:
 - + Input: start date, end date
 - + Output: list of CustomerStat
 - + Assign to class: CustomerStat
- System displays the order list interface of the customer => need a class: ListOrderView:
 - + List of orders of the customer: outsubListOrder
- An order list of the customer is displayed => need a method: `getListOrder()`:
 - + Input: Customer id, start date, end date
 - + Output: list of orders
 - + Assign to class: Order
- System displays an interface contains a list of dishes and combos that was ordered: need a class: OrderStatView:
 - + A list of dishes and combos: outListDishCombo
- When select an order, the system displays a list of dishes and combos => need a method: `getListDishAndCombo()`:
 - + Input: Order id
 - + Output: Order
 - + Assign to class: Order



b. State diagram:

- From ManagementHomeView, if choose view statistic option, the system will display SelectStatisticView
- From SelectStatisticView, if select Customer statistic button, the system will show the CustomerStatView
- From CustomerStatView, after entering start date/ end date and click search, the system will show ListOrderView

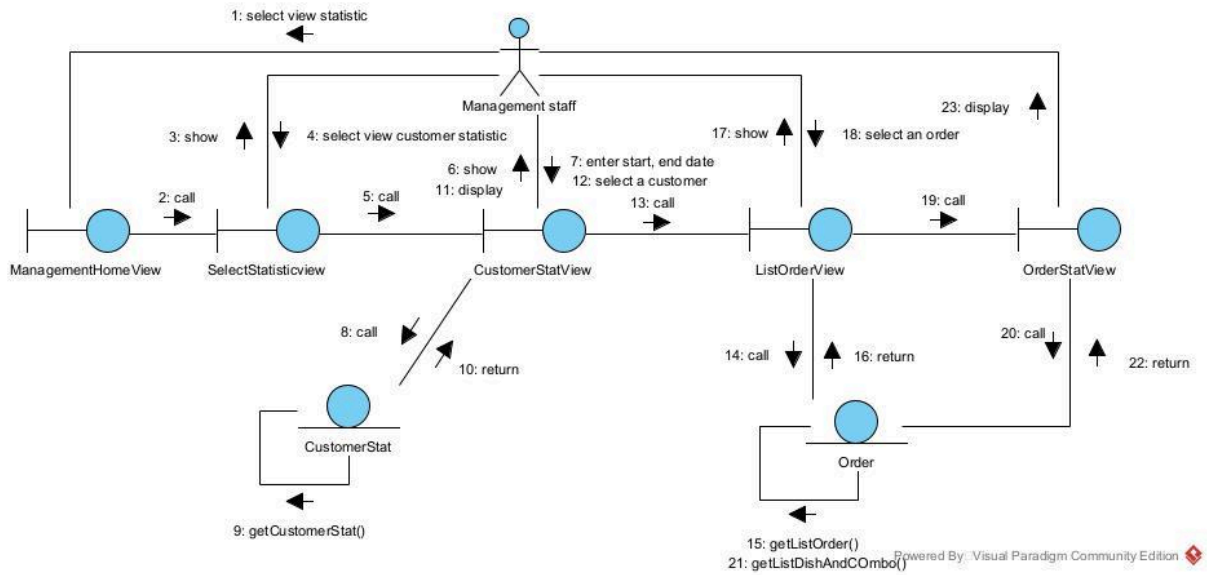
- From ListOrderView, after select an order to view, the system will show OrderStateView and finish



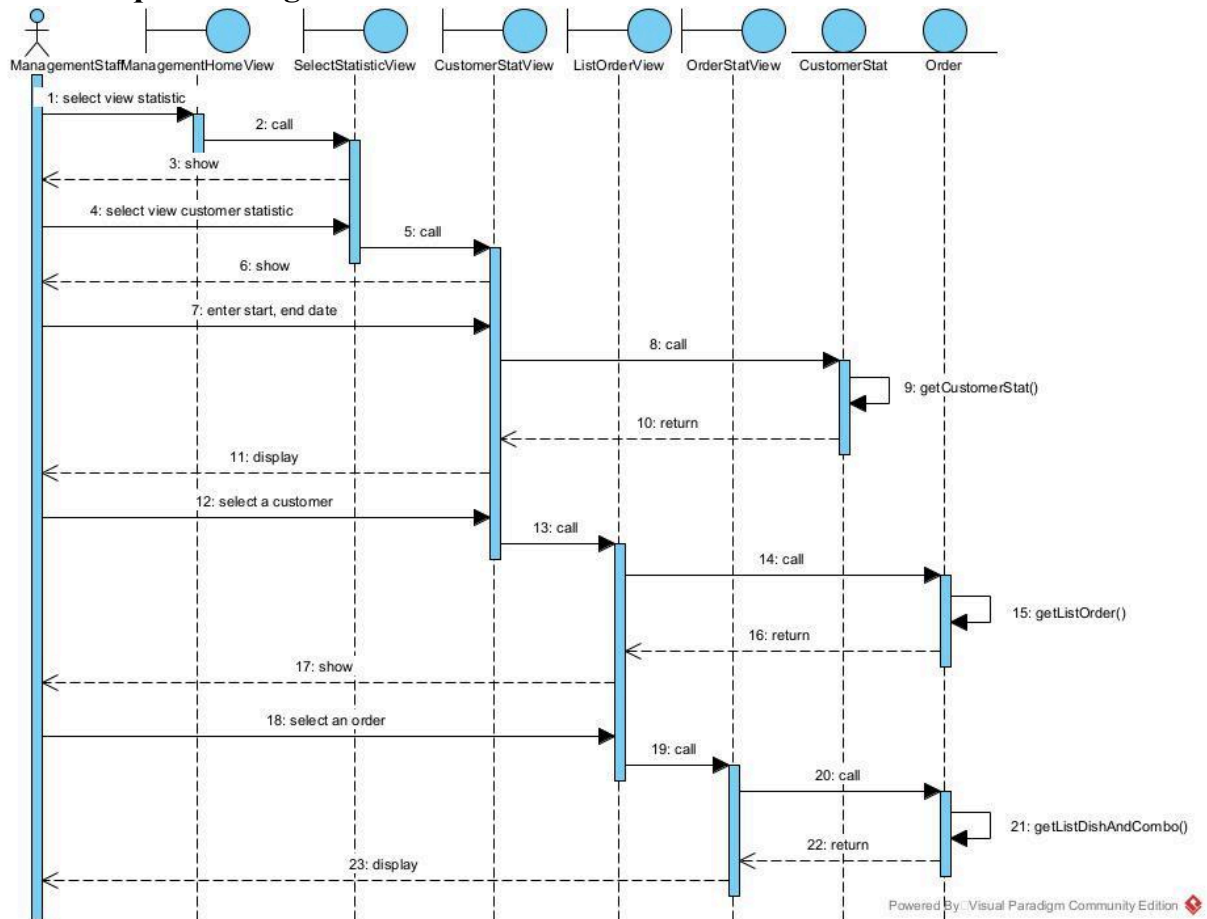
c. Scenario ver 2:

1. The management staff selects the View statistics button
2. The class ManagementHomeView calls the class SelectStatisticView
3. The class SelectStatisticView shows itself to the management staff
4. The management staff clicks on View customer statistics button
5. The class SelectStatisticView calls the class CustomerStatView
6. The class CustomerStatView shows itself to the management staff
7. The management staff enters start date, end date and click search
8. The class CustomerStatView calls the class CustomerStat to process
9. The class CustomerStat calls the method getCustomerStat()
10. The class CustomerStat returns the result to the class CustomerStatView
11. The class CustomerStatView displays the results to the management staff
12. The management staff select a customer
13. The class CustomerStatView calls the class ListOrderView
14. The class ListOrderView calls the class Order to get data
15. The class Order calls the method getListOrder()
16. The class Order returns the results to the class ListOrderView
17. The class ListOrderView shows the results to the management staff
18. The management staff selects an order
19. The class ListOrderView calls the class OrderStatView
20. The class OrderStatView calls the class Order to get data
21. The class Order calls the method getListDishAndCombo()
22. The class Order returns the results to the class OrderStatView
23. The class OrderStatView displays the results to the management staff

d. Communication diagram



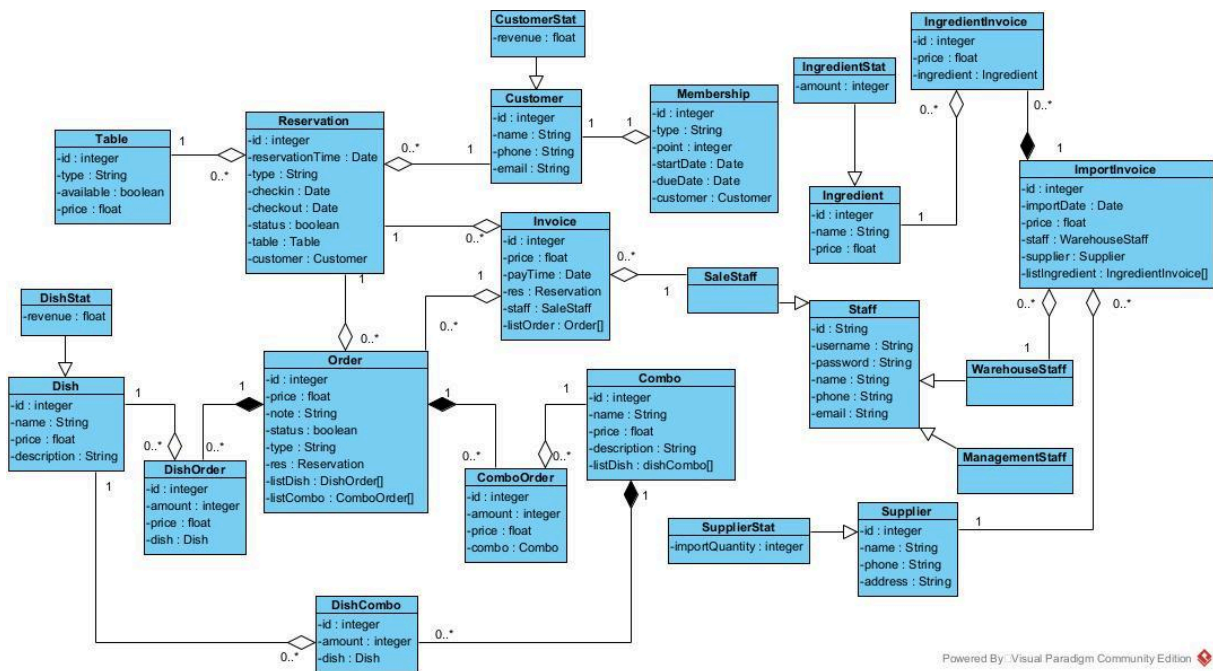
e. Sequence diagram:



III. Design

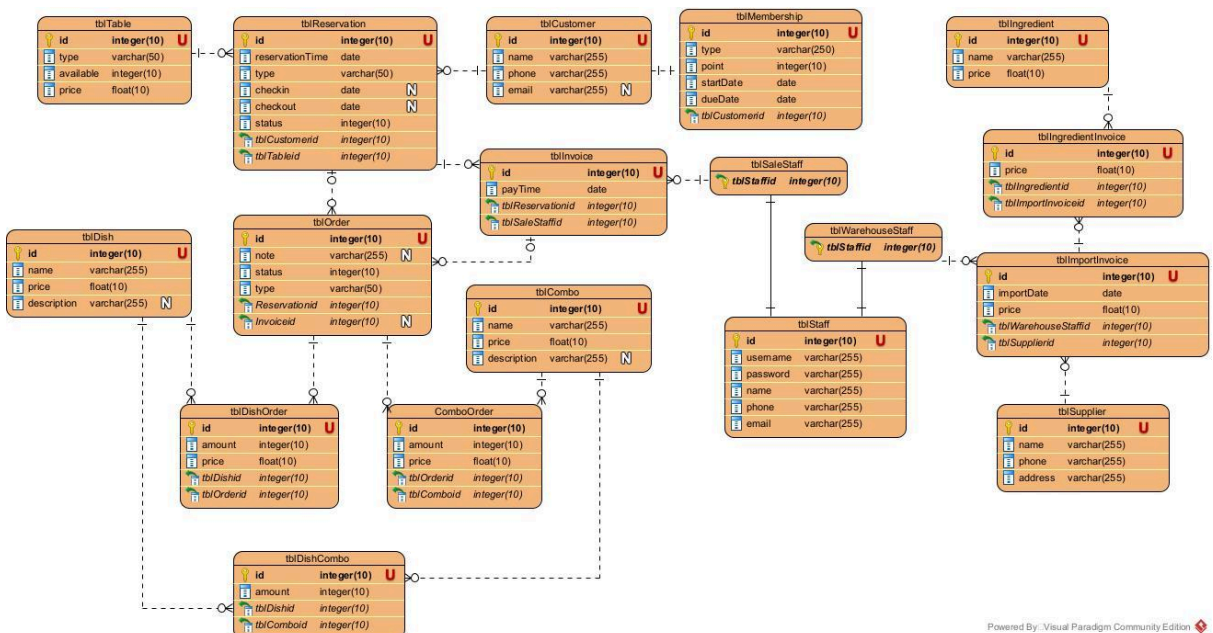
1. Entity class diagram

- Step 1: Add id attribute
- Step 2: Add type of attribute according to Java
- Step 3: Change association into composition and aggregation relationship



2. Database design

- Step 1: Name the table (example: tblTable, tblCustomer)
- Step 2: Take original attribute only in entity class diagram and change the type to SQL type
- Step 3: Consider the relationship between tables (1-1, 1-n)
- Step 4: Add primary key, foreign key
- Step 5: Remove redundant attributes

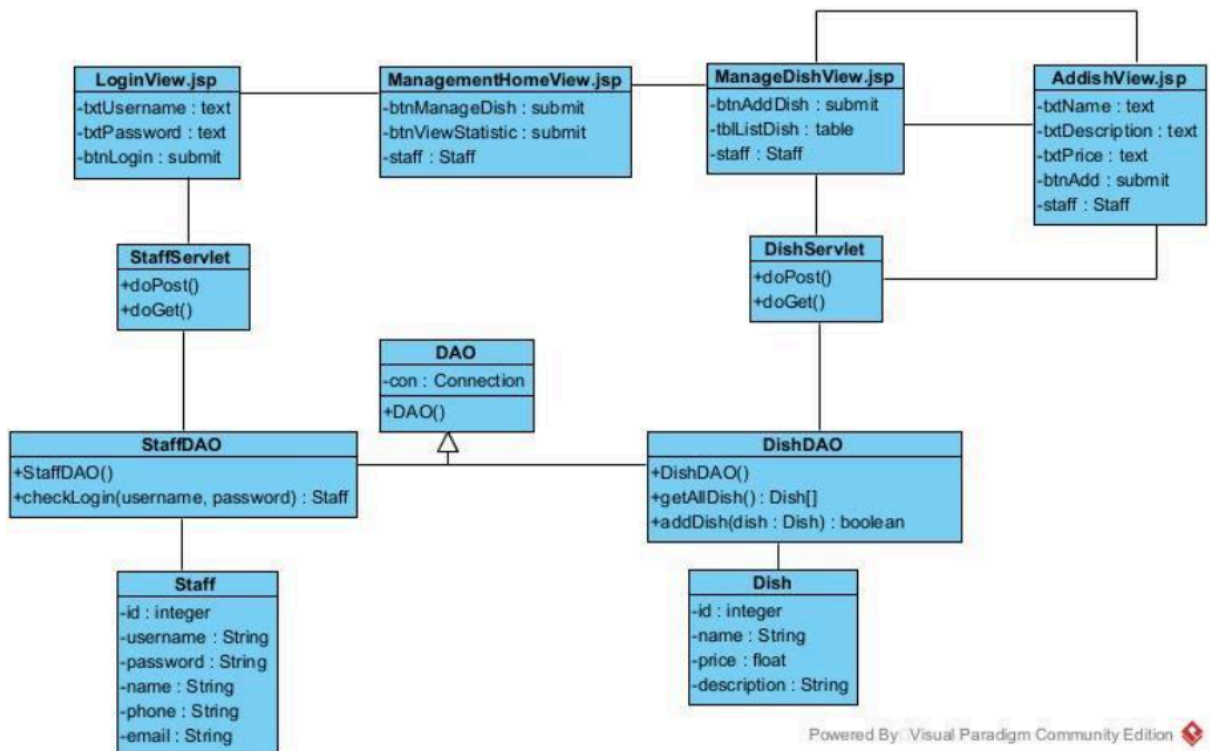
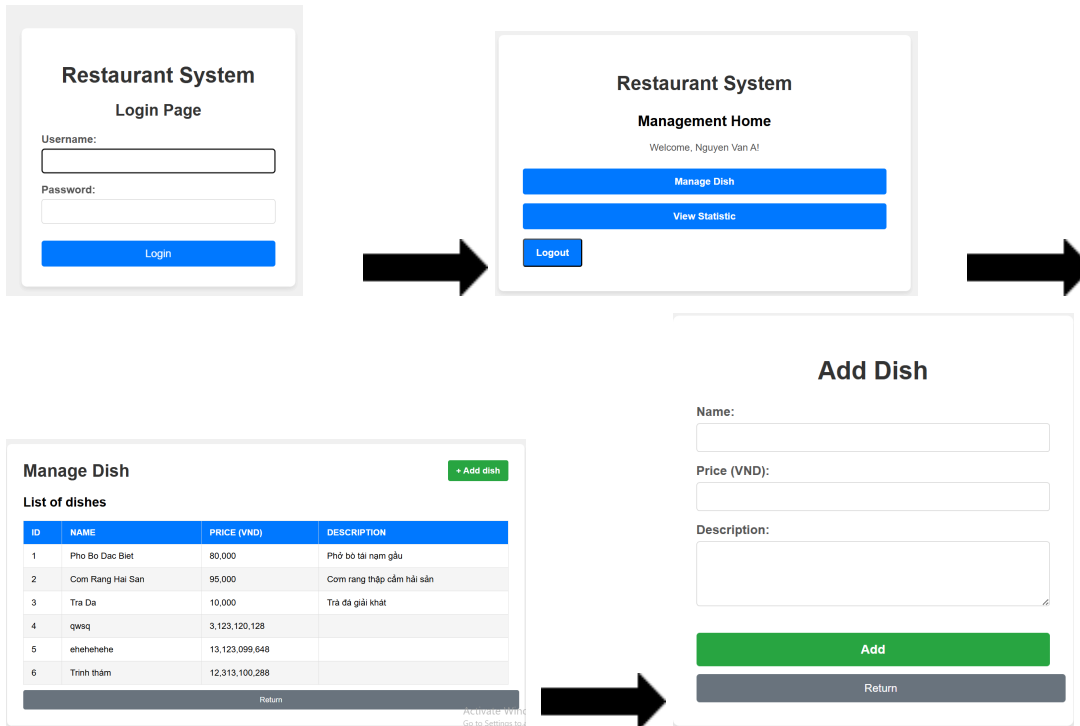


3. Module 1:

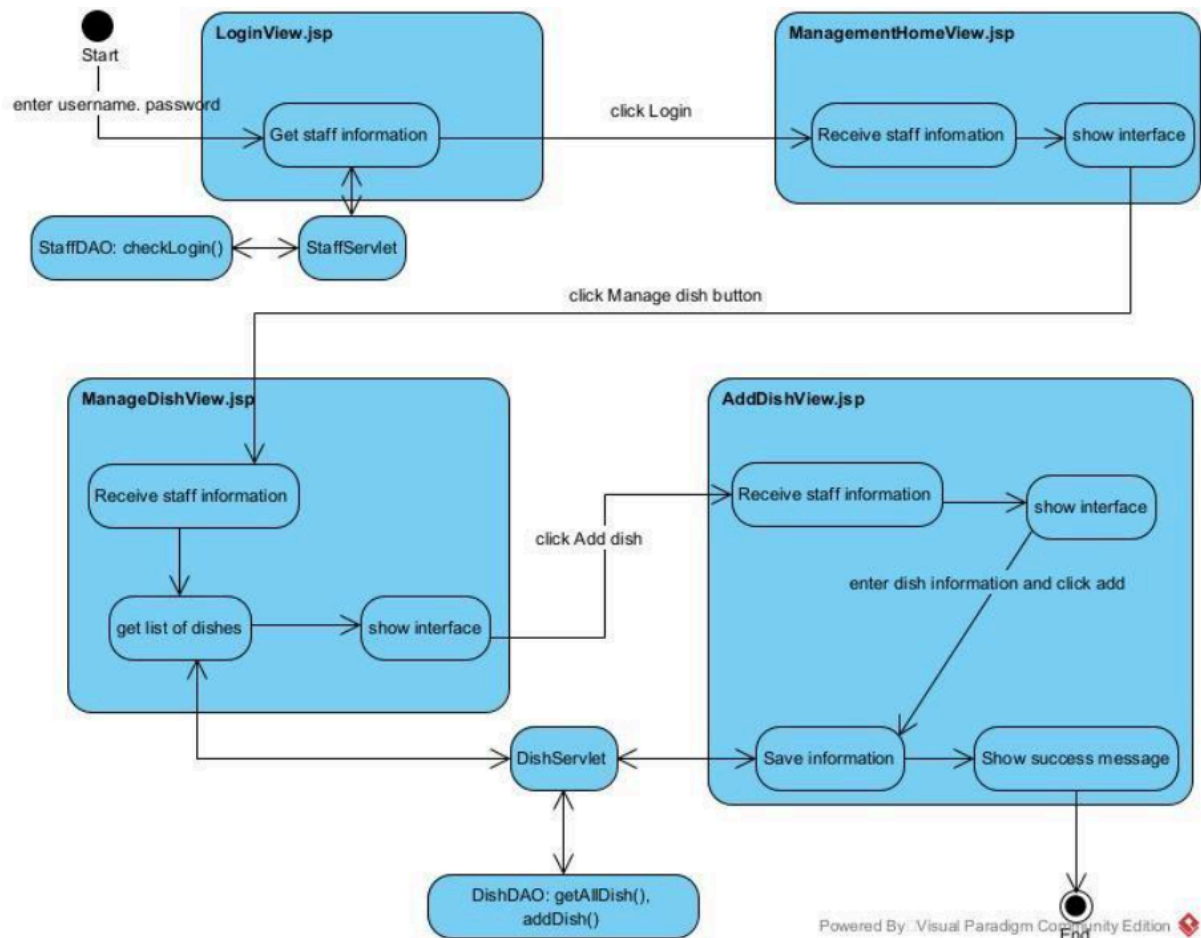
a. Class diagram

- Step 1: Design UI
- Step 2: Create view class
- Step 3: Create DAO class & Servlet class

- Step 4: Adding entity attribute related to modul



b. Activity diagram

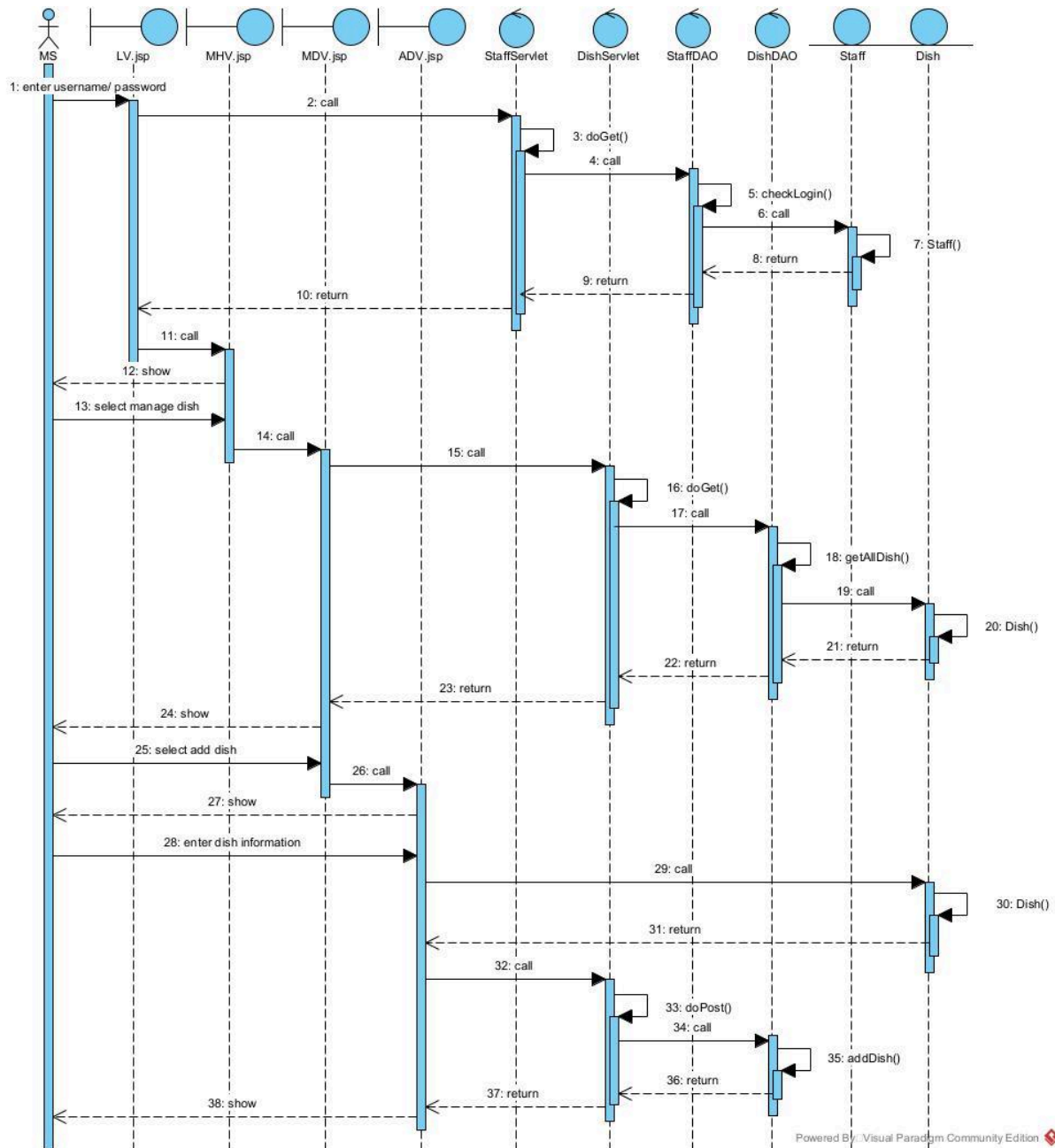


c. Scenario v3

1. A management staff enters username, password and clicks on the login button on LoginView.jsp
2. The interface LoginView.jsp calls StaffServlet
3. The class StaffServlet calls the method doGet()
4. The method doGet() calls the class StaffDAO
5. The class StaffDAO calls the method checkLogin() to check login information
6. The method checkLogin() calls the class Staff
7. The class Staff packs the information to Staff object
8. The class Staff returns the object to the method checkLogin()
9. The method checkLogin() returns the result to the method doGet()
10. The method doGet() returns the result to the interface LoginView.jsp
11. The interface LoginView.jsp calls the interface ManagementHomeView.jsp
12. The interface ManagementHomeView.jsp shows itself to the management staff
13. The management staff selects the the manage dish button
14. The interface ManagementHomeView.jsp calls the interface ManageDishView.jsp
15. The interface ManageDishView.jsp calls the class DishServlet

16. The class DishServlet calls the method doGet()
17. The method doGet() calls the class DishDAO
18. The class DishDAO calls the method getAllDish()
19. The method getAllDish() calls the class Dish
20. The class Dish packs each Dish object
21. The class Dish returns the list of objects to the method getAllDish()
22. The method getAllDish() returns the result to the method doGet()
23. The method doGet() return the result to the interface ManageDishView.jsp
24. The interface ManageDishView.jsp shows itself to the management staff
25. The management staff select add dish button
26. The interface ManageDishView.jsp calls the interface AddDishView.jsp
27. The interface AddDishView.jsp shows itself to the management staff
28. The management staff enters the dish information they want to add and click Add button
29. The interface AddDishView.jsp calls the class Dish
30. The class Dish packs the information into an object
31. The class Dish returns the result to the interface AddDishView.jsp
32. The interface AddDishView.jsp calls the class DishServlet
33. The class DishServlet calls the method doPost()
34. The method doPost() calls the class DishDAO
35. The class DishDAO calls the method addDish()
36. The method addDish() return the result to the method doPost()
37. The method doPost() return the result to the interface AddDishView.jsp
38. The interface AddDishView.jsp show the successful result to the management staff

d. Sequence diagram:



4. Module 2

a. Class diagram

- Step 1: Design UI
- Step 2: Create view class
- Step 3: Create DAO class & Servlet class
- Step 4: Adding entity attribute related to modul



Customer Statistic

Start date (dd/MM/yyyy): End date (dd/MM/yyyy): Search

List of Customer

ID	NAME	PHONE	EMAIL	REVENUE
2	Le Van D	0987654321	d.le@mail.com	815,000 VND
1	Pham Minh C	0912345678	c.pham@mail.com	430,000 VND
3	Vo Thi E	0955555555	e.vo@mail.com	75,000 VND

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List of Orders

ID	TYPE	PRICE (VND)	NOTE
2	In-house	815,000	Khách D, Order 2: Món ăn riêng

[Return](#)

Order Stat Page

Order Id #2

Type: In-house
Note: Khách D, Order 2: Món ăn riêng
Price: 815,000 VND

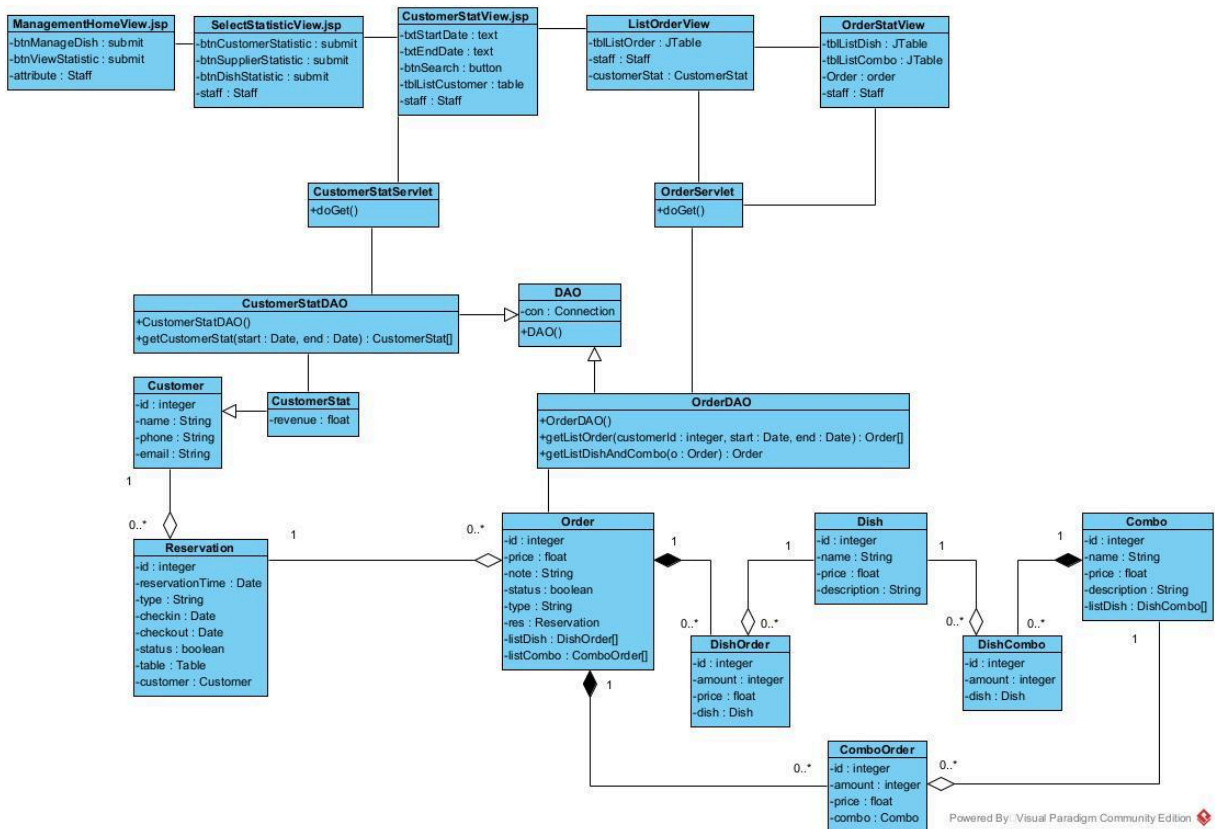
Dish list

Name	Amount	Price	Total
Com Rang Hai San	1	95,000 VND	95,000 VND
Pho Bo Dac Biet	3	240,000 VND	720,000 VND

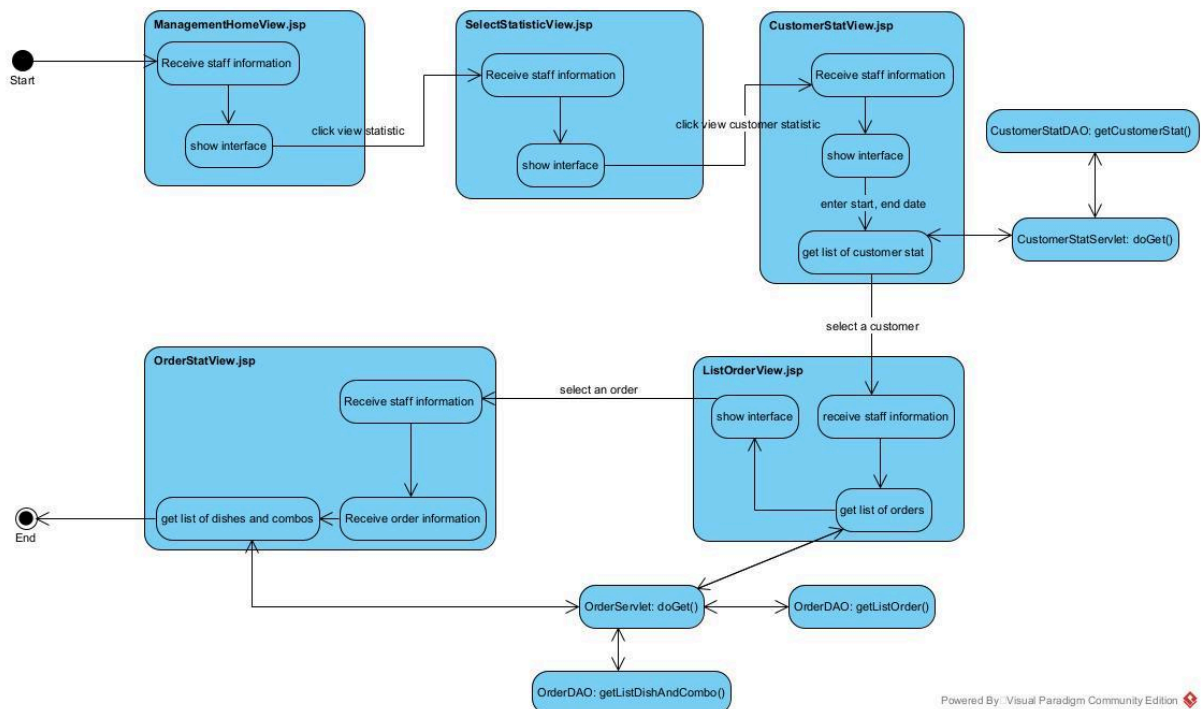
Combo list

No Combo found.

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f. Activity diagram



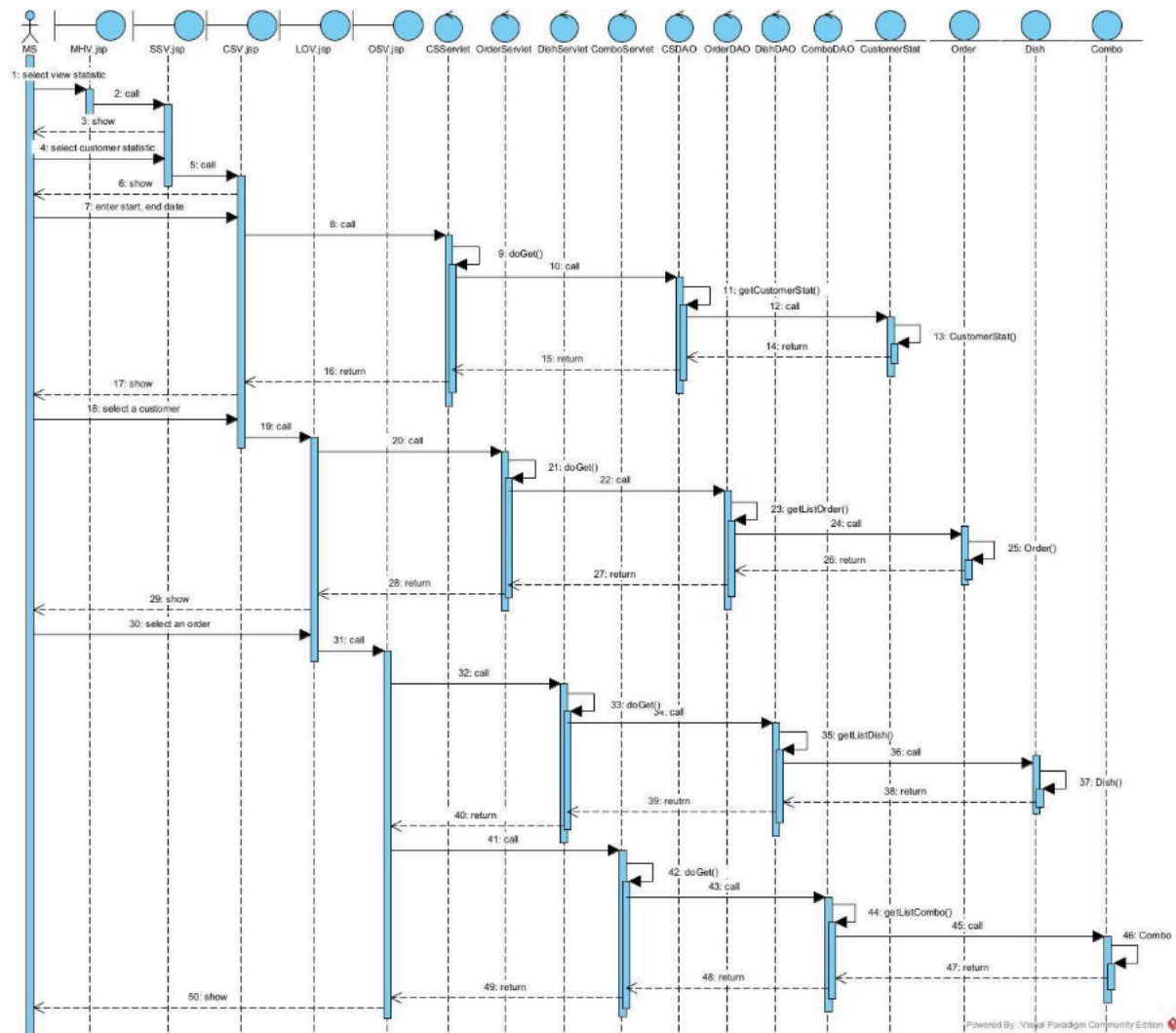
g. Scenario v3

1. The management staff selects the view statistic button in the interface ManagementHomeView.jsp
2. The interface ManagementHomeView.jsp calls the interface SelectStatisticView.jsp
3. The interface SelectStatisticView.jsp shows itself to the management staff

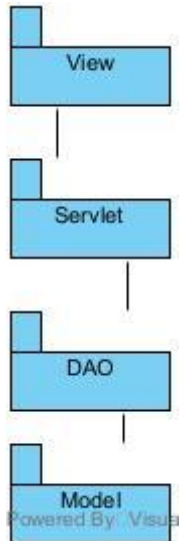
4. The management staff selects the customer statistic button
5. The interface SelectStatisticView.jsp calls the interface CustomerStatView.jsp
6. The interface CustomerStatView.jsp shows itself to the management staff
7. The management staff enters start, end date and click search
8. The interface CustomerStatView.jsp calls the class CustomerStatServlet
9. The class CustomerStatServlet calls the method doGet()
10. The method doGet() calls the class CustomerStatDAO
11. The class CustomerStatDAO calls the method getCustomerStat()
12. The method getCustomerStat() calls the class CustomerStat
13. The class CustomerStat packs the information into an object
14. The class CustomerStat returns the object to the method getCustomerStat()
15. The method getCustomerStat() returns the result to the method doGet()
16. The method doGet() returns the result to the interface CustomerStatView.jsp
17. The interface CustomerStatView.jsp shows the result to the management staff
18. The management staff selects a customer in the list
19. The interface CustomerStatView.jsp calls the interface ListOrderView.jsp
20. The interface ListOrderView.jsp calls the class OrderServlet
21. The class OrderServlet calls the method doGet()
22. The method doGet() calls the class OrderDAO
23. The class OrderDAO calls the method getListOrder()
24. The method getListOrder() calls the class Order
25. The class Order packs each result into an object
26. The class Order returns the objects to the method getListOrder()
27. The method getListOrder() returns the result to the method doGet()
28. The method doGet() return the result to the interface ListOrderView.jsp
29. The interface ListOrderView.jsp shows the result to the management staff
30. The management staff selects an order
31. The interface ListOrderView.jsp calls the interface OrderStatView.jsp
32. The interface OrderStatView.jsp calls the class OrderServlet
33. The class OrderServlet calls the method doGet()
34. The method doGet() calls the class OrderDAO
35. The class OrderDAO calls the method getListDishAndCombo()
36. The method getListDishAndCombo() calls the class DishOrder
37. The class DishOrder packs its normal attributes
38. The class DishOrder call the class Dish
39. The class Dish packs its attributes
40. The class Dish returns the objects to the class DishOrder
41. The class DishOrder return the objects to the method getListDishAndCombo()
42. The method getListDishAndCombo() calls the class ComboOrder

43. The class ComboOrder packs its normal attributes
44. The class ComboOrder call the class Combo
45. The class Combo packs its attributes
46. The class Combo returns the objects to the class ComboOrder
47. The class ComboOrder returns the objects to the method
getListDishAndCombo()
48. The method getListDishAndCombo() returns the result to the method
doGet()
49. The method doGet() returns the result to the interface OrderStatView.jsp
50. The interface OrderStatView.jsp shows the result to the management staff

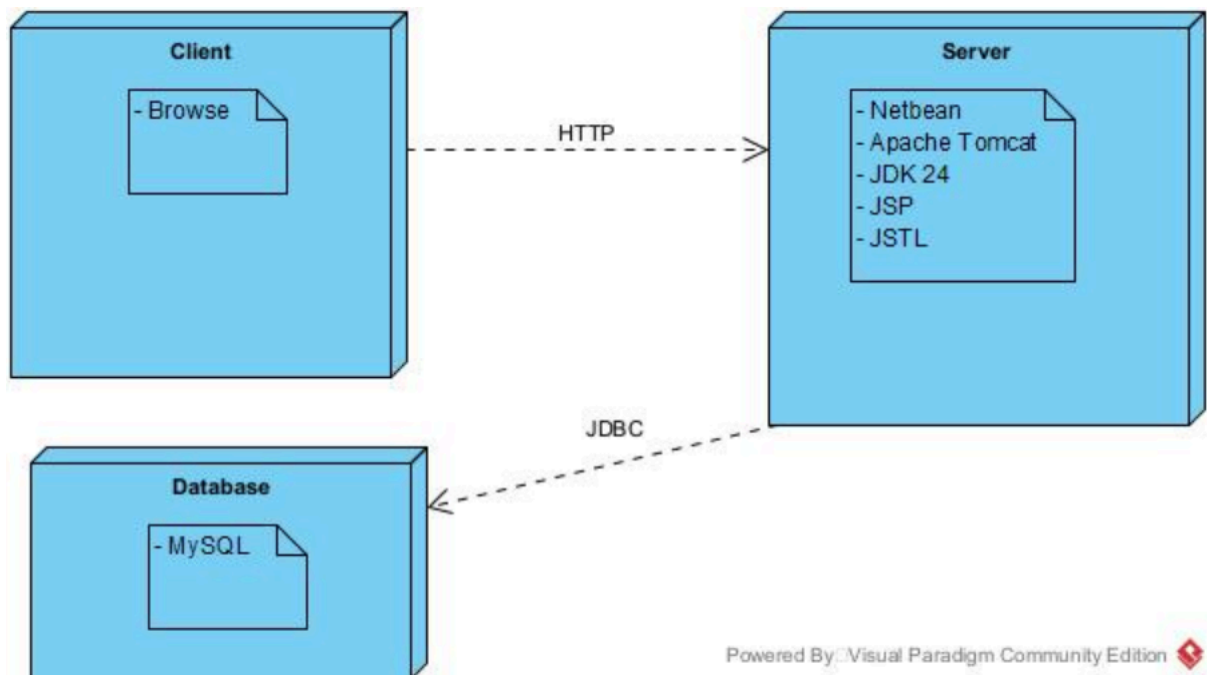
d. Sequence diagram



5. Package diagram



6. Deployment diagram



IV. Programming

1. UI

a. LoginView.jsp

Restaurant System

Login Page

Username:

Password:

Login

b. ManagementHomeView.jsp

Restaurant System

Management Home

Welcome, Nguyen Van A!

Manage Dish

View Statistic

Logout

c. ManageDishView.jsp

Manage Dish

+ Add dish

List of dishes

ID	NAME	PRICE (VND)	DESCRIPTION
1	Pho Bo Dac Biet	80,000	Phở bò tái nạm gầu
2	Com Rang Hai San	95,000	Cơm rang thập cẩm hải sản
3	Tra Da	10,000	Trà đá giải khát
4	qwsq	3,123,120,128	

Return

d. AddDishView.jsp

Add Dish

Name:

Price (VND):

Description:

Add

Return

e. SelectStatisticView.jsp

Restaurant system

Select Statistic

Customer Statistic

Dish Statistic

Supplier Statistic

Return

f. CustomerStatView.jsp

Customer Statistic

Start date (dd/MM/yyyy):

10/01/2025

End date (dd/MM/yyyy):

10/10/2025

Search

List of Customer

ID	NAME	PHONE	EMAIL	REVENUE
2	Le Van D	0987654321	d.le@mail.com	815,000 VND
1	Pham Minh C	0912345678	c.pham@mail.com	430,000 VND
3	Vo Thi E	0955555555	e.vo@mail.com	75,000 VND

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g. ListOrderView.jsp

List of Orders

ID	TYPE	PRICE (VND)	NOTE
2	In-house	815,000	Khách D, Order 2: Món ăn riêng

Return

h. OrderStatView.jsp

Order Stat Page

Order Id #2

Type: In-house
Note: Khách D, Order 2: Món ăn riêng
Price: 815,000 VND

Dish list

Name	Amount	Price	Total
Cơm Rang Hai San	1	95,000 VND	95,000 VND
Pho Bo Dac Biet	3	240,000 VND	720,000 VND

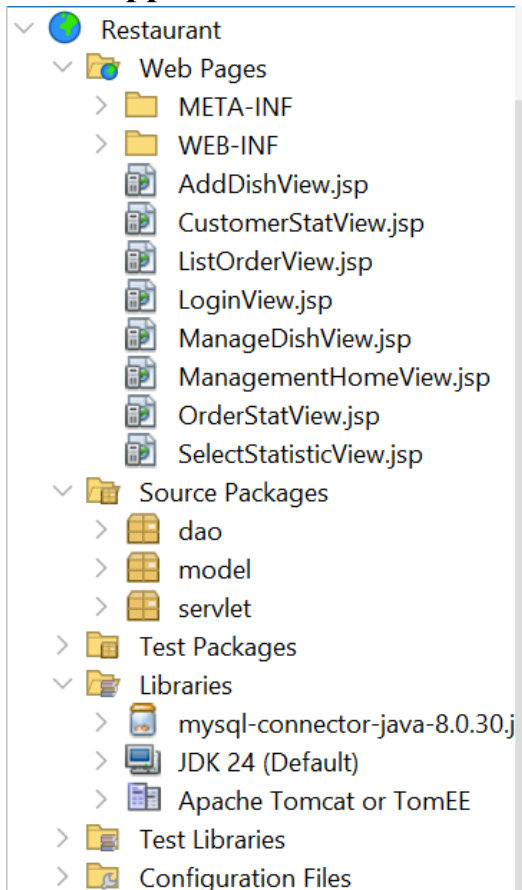
Combo list

No Combo found.

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2. App structure



3. Link source code

<https://github.com/NTM-2004/PTTK-HTTP-24-Restaurant>